

Daniel DeKerlegand

AI Researcher & Senior Engineer

New Orleans, LA

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SUMMARY

Senior engineer and PhD candidate in computer science with a concentration in AI/ML. 10+ years of professional software experience in AI/ML, API, web, mobile, and distributed systems engineering. Doctoral research assistant conducting research in AI/ML and NLP with a focus on hybrid and neuro-symbolic AI architectures, procedural generation, and social physics.

EXPERIENCE

Evolve Media AI

Jun 2021 – Present

Director of Artificial Intelligence · New Orleans, LA

- Architected production microservices platform with 8 services (Python Flask, Node.js Express, React) deployed on Kubernetes (AWS EKS, GKE, AKS) using Infrastructure-as-Code (Terraform), implementing GraphQL API, WebSocket subscriptions, and Redis pub/sub for real-time collaborative applications serving thousands of users.
- Developed sophisticated AI orchestration system using Google Gemini 2.5 Pro with 14+ function calling tools for multi-modal content analysis (video, audio, text), implementing machine learning preference systems with weighted scoring algorithms for personalized recommendations and scalable video intelligence pipelines with parallel processing architectures.
- Built comprehensive full-stack applications including real-time video editor with React/Remotion, cross-platform mobile app (React Native/Expo) with biometric authentication, and REST/GraphQL APIs with JWT authentication, PostgreSQL database, automated deployment pipelines, and monitoring achieving 99.9% uptime.
- Designed intuitive AI-powered interfaces with conversational assistants, multi-turn dialogue support, real-time progress feedback, and personalized user experiences combining symbolic reasoning with neural generation for adaptive content creation responding to user behavior patterns.

- Conducted a multi-phase doctoral research program — dissertation: Insimul: A Hybrid AI System for Social Immersion — spanning five implemented systems across historical education, pedagogy, entertainment, and language learning, establishing concrete architectural patterns for combining symbolic social physics with statistical language generation and virtual reality.
- Built Insimul, a neuro-symbolic narrative AI infrastructure unifying three previously independent social simulation systems (Ensemble, Talk of the Town, Kismet) behind a single declarative Prolog format, executed by tau-prolog in the browser with SWI-Prolog as server fallback, alongside a provider-agnostic LLM interface, procedural world and character generation, and AI-assisted authoring that generates Prolog predicates from natural-language descriptions.
- Engineered the full stack in TypeScript: a React collaborative authoring client and a Node/Express server exposing REST, WebSocket, Server-Sent Events, and gRPC surfaces, with editor content in PostgreSQL via Drizzle ORM mirrored into MongoDB for the runtime, and a two-phase architecture separating procedural generation at creation time from simulation at execution time.
- Closed the symbolic–statistical loop in LLVR, a VR language-learning deployment in which socio-linguistic rules, quest structure, and grammar constraints condition speech recognition and dialogue generation, while learner speech continuously updates the knowledge base and truth layer and is reflected back as NPC behavior; evaluated through expert assessment by professional domain specialists and computational trace analysis.
- Established a methodology for collaborative authoring with domain experts on VESPACE, a VR reconstruction of 18th-century Parisian theatre culture, in which literary historians authored 230 social physics rules under a "citation as AI" practice grounding every rule in textual evidence — making interpretive assumptions transparent and auditable.

Torsh

Dec 2015 – Jun 2021

Lead Engineer · New Orleans, LA

- Oversaw, from conception to production, the complete rewrite of legacy iOS and Android apps as a single React Native app. Directed other engineers and operationalized organizational engineering standards.
- Architected and implemented app navigation, authentication, data persistence, and core UI components. Built bespoke React Native libraries in Objective C and Java for audio/video domain-specific tasks like Bluetooth connectivity and camera management.
- Designed, developed, and deployed GraphQL API in Node.js. Integrated Apollo client into application, adding an offline-first local caching system.
- Implemented critical fullstack features in a Meteor/Node.js web application: a pub-sub task queuing system, audio-video processing services, an internal reporting library of data analytics, a PDF annotation tool, an OpenWRT video chat tool, and a CMS.

EDUCATION

University of New Orleans New Orleans, LA

Doctor of Philosophy, Computer Science Expected May 2026

Master of Science, Computer Science 2018

Bachelor of Arts, English Language & Literature · Bachelor of Arts, Film/Cinema/Video Studies 2010

University of Arizona Tucson, AZ

Master of Fine Arts, Creative Writing 2013

CORE SKILLS

Symbolic AI & Logic

SWI-Prolog Ensemble Insimul DSL Datalog ASP CLIPS Pyke
Knowledge Graphs OWL SPARQL Social Physics

AI/ML Tooling

PyTorch Keras TensorFlow MLFlow Jupyter HuggingFace

LLM Inference, Training & Optimization

Open R1 MinGPT Megatron LM Finetune Transformer LM DSPy
Retrieval-Augmented Generation FAISS ChromaDB Pinecone
Vector Embeddings Semantic Search Prompt Engineering
Few-Shot Learning Chain-of-Thought Prompting LoRA/ControlNet

Agentic AI

- Model Context Protocol
- Agent-to-Agent
- Agent Development Kit
- Google Vertex Agents
- LangChain
- LangGraph
- CrewAI
- AutoGen
- BabyAGI
- AgentGPT
- Semantic Kernel
- ReAct
- Dify
- Flowise

Speech & Audio

- ffmpeg
- Whisper STT
- Coqui TTS
- Piper TTS
- ElevenLabs
- Google Cloud TTS
- espeak-ng
- Oculus Lip Sync
- SALSA
- Azure Speech Services
- Amazon Polly
- Wav2Vec
- DeepSpeech
- Tacotron
- VITS
- Phoneme Analysis

Game Engines & VR

- Unity
- Unity Sentis
- Unreal Engine 5
- MetaHumans
- Oculus SDK
- OpenXR
- SteamVR
- XR Interaction Toolkit
- Godot
- WebXR
- A-Frame
- Spatial Audio SDK

Statistical Model Architectures

- RNN
- CNN
- LSTM
- GAN
- BERT
- Transformer
- T5
- Attention Mechanisms
- Encoder-Decoder Models
- VAE
- ResNet
- U-Net
- Graph Neural Networks
- Diffusion Models

Procedural Generation

- Tracery
- Perlin Noise
- Wave Function Collapse
- L-Systems
- Markov Chains
- Context-Free Grammars
- Cellular Automata
- Noise Functions (Simplex, Worley)
- PCG Algorithms
- Rule-Based Generation

Data Analysis & Research

- R
- lme4
- Montreal Forced Aligner
- NVivo
- Python (NumPy, Pandas, SciPy)
- Matplotlib
- ggplot2
- SPSS
- Jupyter Notebooks
- Tableau
- Statistical Modeling
- Mixed-Effects Models

Languages & Frameworks

- Python
- Rust
- Go
- TypeScript
- Node.js
- C#/.NET
- Clojure
- Java
- React
- Vue
- React Native
- Swift
- Kotlin

DevOps/MLOps, Data & Cloud-Native

- Terraform
- Helm
- Chef
- Ansible
- Kubernetes
- Jenkins
- Jenkins X
- CircleCI
- AWS
- GCP
- Grafana
- Sentry
- Datadog
- Kibana
- Prometheus
- GitHub Actions
- MongoDB
- PostgreSQL
- Drizzle ORM
- MySQL
- SQLite
- Redis
- Neo4j
- Prisma
- TypeORM
- Supabase